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B.TECH
(SEM III) THEORY EXAMINATION 2022-23
HEAT TRANSFER OPERATIONS

Time: 3 Hours

Total Marks: 100

Note: Attempt all sections. If require any missing data; then choose suitably.

SECTION – A

1. Attempt all questions in brief.

2×10=20

- (a) What are the units of thermal conductivity and heat transfer coefficient?
- (b) Define insulation.
- (c) State the Newton's law of cooling.
- (d) Explain forced convection?
- (e) State the Stephan Boltzmann law for radiation.
- (f) Define steam economy.
- (g) Define Radiation intensity.
- (h) Write the physical significance of Nu and Pr numbers.
- (i) What is black body?
- (j) Give the classification of heat exchanger.

SECTION – B

2. Attempt any three of the following:

10×3=30

- (a) Define critical thickness of insulation. Derive an expression for critical thickness of insulation of cylinder.
- (b) An industrial wall is constructed of 20cm thick fireclay, $K=1\text{W/m}^0\text{C}$. This is covered on outer surface with a 3 cm layer of insulating material, $K=0.075\text{ W/m}^0\text{C}$. The innermost surface is at 94^0C & the outermost at 40^0C . Calculate the steady rate of heat transfer through the wall in W/m^2 & the temperature of the interface between the fireclay & the insulating material.
- (c) Derive an expression for general three dimensional heat conduction equation for general three dimensional heat conduction equation in cylindrical coordinate.
- (d) What is importance of LMTD? Derive the equation for parallel flow arrangement.
- (e) Derive heat transfer equation for a cylindrical vessel used for heat exchanger.

SECTION -C

3. Attempt any one part of the following:

10×1=10

- (a) Derive an expression for heat conduction through a spherical wall.
- (b) Explain the lumped heat capacity method for unsteady state heat conduction.

4. Attempt any one part of the following:

10×1=10

- (a) The outside surface temperature of a pipe ($r=0.1$ m) is 400 K. The pipe is losing heat to atmosphere, which is at 300K. The film heat transfer coefficient is $10 \text{ W/m}^2\text{K}$. To reduce rate of heat loss, the pipe is insulated by a 50 mm thick layer of asbestos, $k=0.5\text{W/mK}$. Calculate the % reduction in the rate of heat loss.
- (b) What are Fins? Describe the different kinds and their applications.

5. Attempt any one part of the following:

10×1=10

- (a) Derive an expression for instantaneous temperature and heat transfer rate for a body subjected to heating or cooling in terms of Biot and Fourier number.
- (b) Obtain the rate of heat loss from a thermoflask if the polished silvered surfaces have an emissivity of 0.05, the liquid in the flask is at 95°C and the casing is at 20°C . Calculate the loss if both surfaces were black. Stefan- Boltzmann constant = $4.875 \times 10^{-8} \text{ kcal/hr m}^2 \text{ K}^4$.

6. Attempt any one part of the following:

10×1=10

- (a) What do you mean by hydrodynamic and thermal boundary layer? Explain with a neat sketch.
- (b) What is Condensation? Why it is difficult to achieve and maintain the dropwise condensation? Also discuss the effect of noncondensable gases on condensation.

7. Attempt any one part of the following:

10×1=10

- (a) Hot oil with a capacity rate of 2500 W/K flows through a double pipe heat exchanger. It enters at 360°C and leaves at 300°C . Cold fluid enters at 30°C and leaves at 200°C . If the overall heat transfer coefficient is $800 \text{ W/m}^2\text{K}$. Determine the heat exchanger area required for (i) parallel flow and (ii) Counter flow.
- (b) Define Kirchhoff's law of radiation and derive the expression for emissivity for radiation between two infinite and parallel plates.